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A MINOR PROJECT REPORT ON
“Digital Forensic Investigation and Evidence Recovery”

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Abstract

Digital forensics and storage media analysis play a pivotal role in contemporary criminal investigations, incident response, and judicial proceedings. The objective of this minor project is to execute a forensically sound, end-to-end investigation methodology encompassing evidence seizure, bit-stream physical acquisition, cryptographic verification, deleted file recovery, file carving, steganography detection, and file signature anomaly analysis. The primary investigation framework evaluated in this report is the CyberCheck 6.0 Forensic Suite (incorporating TrueBackWin 2.1), an enterprise-grade digital forensics environment developed by C-DAC.

The investigation was conducted on a suspect 16 GB SanDisk USB flash device containing a FAT32 file system. Using TrueBackWin 2.1, a raw bit-stream physical image was acquired with simultaneous MD5 and block-level hashing, guaranteeing complete mathematical reproducibility and legal chain of custody compliance. Subsequent analysis within CyberCheck Probe successfully reconstructed deleted partitions, extracted unallocated cluster space, carved corrupted and overwritten documents (PPTX, DOCX, HTML, PDF), rendered hidden visual/multimedia artifacts, verified steganographic payload presence, and flagged header-extension signature mismatches.

Key Forensic Focus Areas

Keywords: Digital Investigation, Computer Forensics, Bit-Stream Imaging, TrueBackWin 2.1, CyberCheck 6.0, Cryptographic Hashing, Data Carving, Steganography Detection, File Signature Mismatch, Unallocated Space Analysis.

1. Comprehensive Introduction & Forensic Standards

Digital forensics is the scientific discipline dedicated to the identification, preservation, extraction, documentation, and analysis of digital evidence stored on electronic media. In legal and corporate inquiries, maintaining the absolute integrity of digital evidence is paramount. Any unauthorized modification, unrecorded access, or failure to document operational procedures can render evidence inadmissible in a court of law.

1.1 Legal Framework & Chain of Custody Guidelines

To adhere to international digital evidence standards (ISO/IEC 27037) and Indian Information Technology Act rules, the acquisition process must observe strict Write-Blocking principles. Physical or software write-blockers must prevent write operations to source media. Furthermore, a rigorous Chain of Custody (CoC) log must record the investigator identity, law enforcement agency, crime reference number, physical media serial numbers, acquisition timestamps, and cryptographic checksums before and after analysis.

1.2 Architectural Overview of CyberCheck 6.0 Suite

CyberCheck 6.0 is an advanced digital forensic solution designed for comprehensive disk analysis, data carving, and incident triage. The suite consists of two core operational modules:

- 1. TrueBackWin 2.1: A dedicated media seizure and bit-stream acquisition tool capable of low-level sector reading, raw physical imaging, split-volume generation, and multi-threaded cryptographic hash calculation (MD5, SHA-1, SHA-256).
- 2. CyberCheck Probe: A sophisticated forensic analysis workspace providing file system parsing, unallocated cluster extraction, automated steganography detection, header-extension signature matching, GREP/keyword searching, and integrated multimedia previewing.

1.3 Hardware & Storage Media Specification

The target evidence evaluated during this project comprised a suspect removable USB flash storage device with the following parameters:

Parameter	Specification / Value
Device Hardware Model	SanDisk Removable USB Storage Device
Hardware Serial Number	4C530001170403112591
Total Capacity & Sectors	14.53 GB / 30,464,000 Total Sectors (512 Bytes/Sector)
File System Architecture	FAT-32 (File Allocation Table 32)
Acquisition Image File	Pd forensics_01.P01 (Split Raw Bit-Stream)

2. Phase I: Seizure, Physical Imaging & Hashing (TrueBackWin 2.1)

The initial operational phase requires securing the physical evidence, establishing write-block protection, and capturing a bit-stream physical image of the suspect storage media. The step-by-step procedure executed in TrueBackWin 2.1 is detailed below.

Step 1: Initializing Seize & Acquire Module

Launch the CyberCheck 6.0 software suite under administrator privileges. Navigate to the main menu bar, select 'Tools', and click on 'Seize & Acquire' from the dropdown list. This initializes the standalone TrueBackWin 2.1 module dedicated to media seizure.

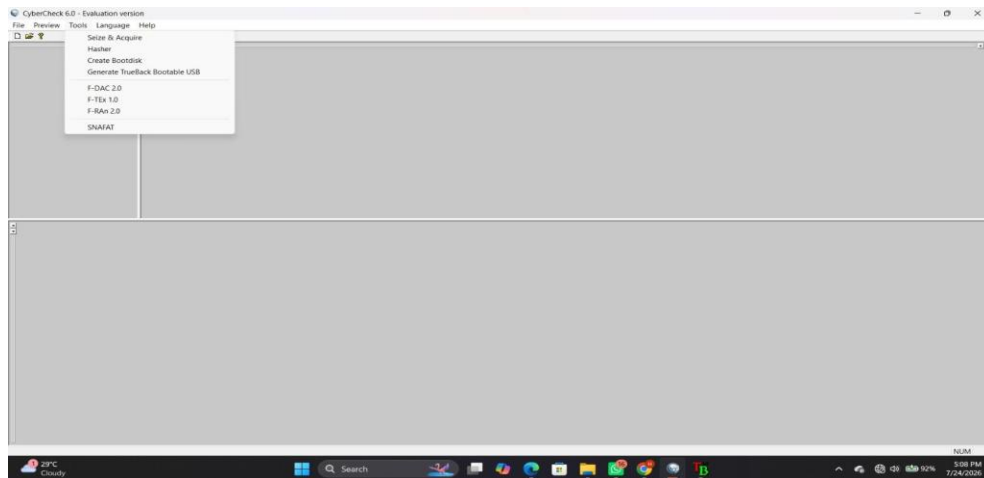


Figure 1.1 — Initial Window of CyberCheck 6.0

Step 2: Selecting Dual Seizure and Acquisition Mode

Upon opening the TrueBackWin 2.1 interface, select the 'Seize & Acquire' option from the splash menu. This mode enforces simultaneous cryptographic hashing and bit-stream disk imaging at the crime scene or laboratory environment.

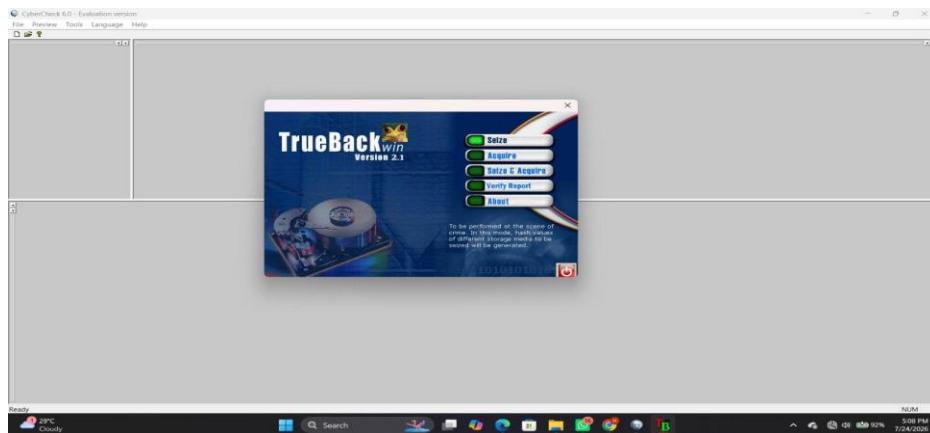


Figure 1.2 — Window of TrueBackWin Tool

Step 3: Formal Case Metadata & Chain of Custody Input

Complete all mandatory fields in the 'Seizure & Acquisition' metadata entry form to establish legal chain of custody. Enter Investigator Name ('group 2'), Rank ('1st'), Police Station ('Mumbai police Station'), Crime Number ('7'), Place of Seizure ('Mumbai'), Suspect Name ('Rahul'), Suspect Addresses ('Juhu mumbai / Nashik'), Witness Names ('Harshal' and 'Akshay'), Lab Reference Number ('123'), and Evidence File Name ('Pd forensics').

The screenshot shows the 'TrueBackWin 2.1 - Seizure & Acquisition' window. The 'Options' tab is active, displaying a form for entering case metadata. The form includes fields for Investigator's Name (group 2), Rank (1st), Police Station (Mumbai police Station), Crime Number (7), Seizure Memo Number, Place of Seizure (Mumbai), Date of Hashing (24/07/2026), Time of Hashing (17:20:21 PM), Name of Suspect (Rahul), Address1 (Juhu mumbai), Address2 (Nashik), Name of Witness 1 (Harshal), Address1 (mumbai), Address2 (mumbai), Notes (Forensics), Lab Reference Number (123), and Evidence File Name (Pd forensics). A note at the bottom states: 'Note: It is the User's responsibility to write protect the media to be Seized & Acquired'. The window has 'Back' and 'Next' buttons at the bottom right.

Figure 1.3 — Case and Seizure Metadata Input Interface

Step 4: Source Device Identification & Target Storage Mapping

Under the Physical Disks pane, identify and select the suspect target drive: 'Disk 1 - Removable SanDisk USB device' (14.53 GB, 30,464,000 sectors, Serial No. 4C530001170403...). Next, designate the secure destination partition (local fixed partition C: with 43.08 GB free space) where the forensic image segments will be stored.

The screenshot shows the 'TrueBackWin 2.1 - Seizure & Acquisition' window with the 'Select Media' tab active. It displays two tables: 'Physical Disks' and 'Destination'. The 'Physical Disks' table has columns for Disk, Media Type, Interface, Status, Size, Sectors, Model No., and Serial No. It lists Disk 0 (Fixed Unknown disk, 931.51 GB, 1953525168 sectors, WD_BLACK SN770 1TB) and Disk 1 (Removable Media, 14.53 GB, 30464000 sectors, SanDisk, 4C530001170403...). The 'Destination' table has columns for Disks, Total Size, Used Size, Free Size, File System, and Disk Type. It lists C: (New Volume, 218.21 GB, 7.49 GB, 210.72 GB, NTFS, Fixed) and D: (New Volume, 218.21 GB, 158.25 GB, 59.96 GB, NTFS, Fixed). A note at the bottom states: 'Note: Press "Refresh" if you have added/removed any media.' The window has 'Refresh', 'Back', and 'Next' buttons at the bottom right.

Figure 1.4 — Source Media and Destination Disk Selection Interface

Step 5: Acquisition Format, Hashing, & Chunking Configuration

Configure imaging settings: Select 'TrueBackWin Image' as the output file format, set the primary Hash Type to 'MD5', leave Compression set to 'None' for optimal acquisition speed, set the image split chunk size to 2 GB with destination path C:\group 2\7\Pd forensics_01.P01, and enable 'Use Block Hash' configured for 15,360 sectors per block across 1,984 total blocks.

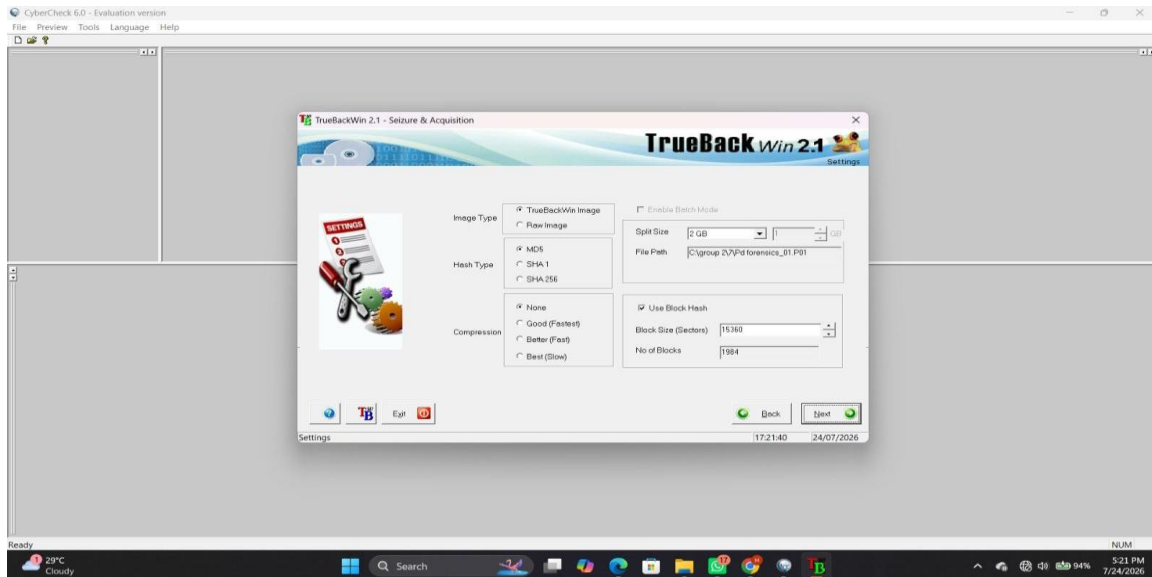


Figure 1.5 — Image Format and Hashing Configuration Settings Interface

Step 6: Final Verification & Confirmation of Seizure Metadata

Review the 'Seizure & Acquisition Case Data Summary' dialog window. Audit all populated metadata, timestamps, and path configurations. Select the radio button 'Confirmed as correct' and click 'Next' to trigger low-level disk imaging.

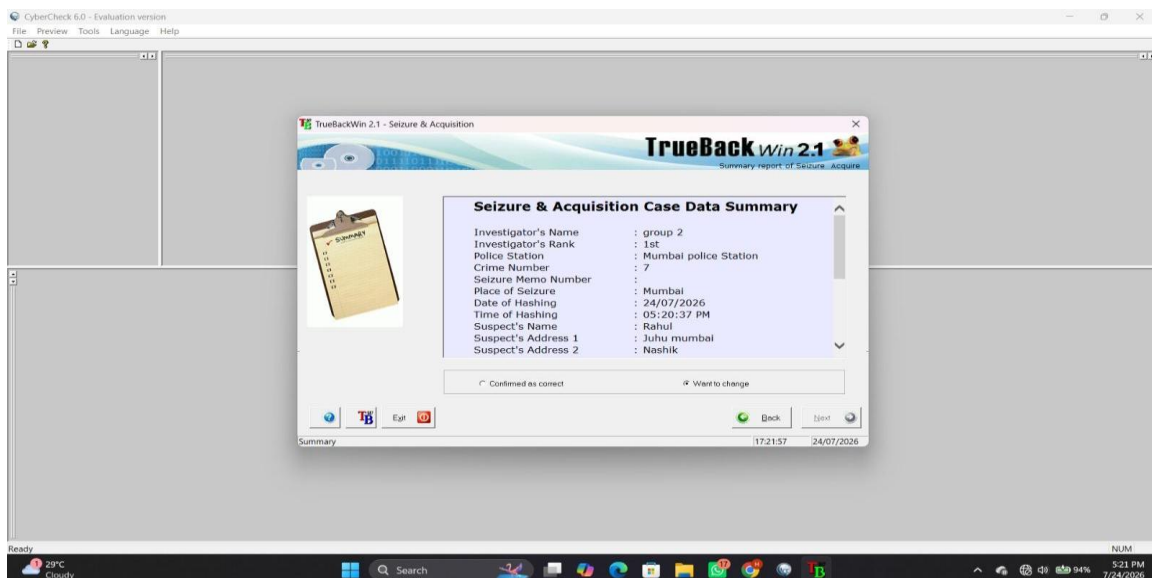


Figure 1.6 — Seizure & Acquisition Case Data Summary Review Window

Step 7: Real-Time Bit-Stream Imaging Execution & Monitoring

Monitor the live imaging progress initialized at 05:22:09 PM. TrueBackWin reads physical disk sectors sequentially, displaying real-time performance metrics: throughput speed (27.22 MB/sec), sector progress (222,976 / 30,463,999 sectors), elapsed time (4 sec), estimated remaining time (9 min 2 sec), and zero bad sectors encountered.

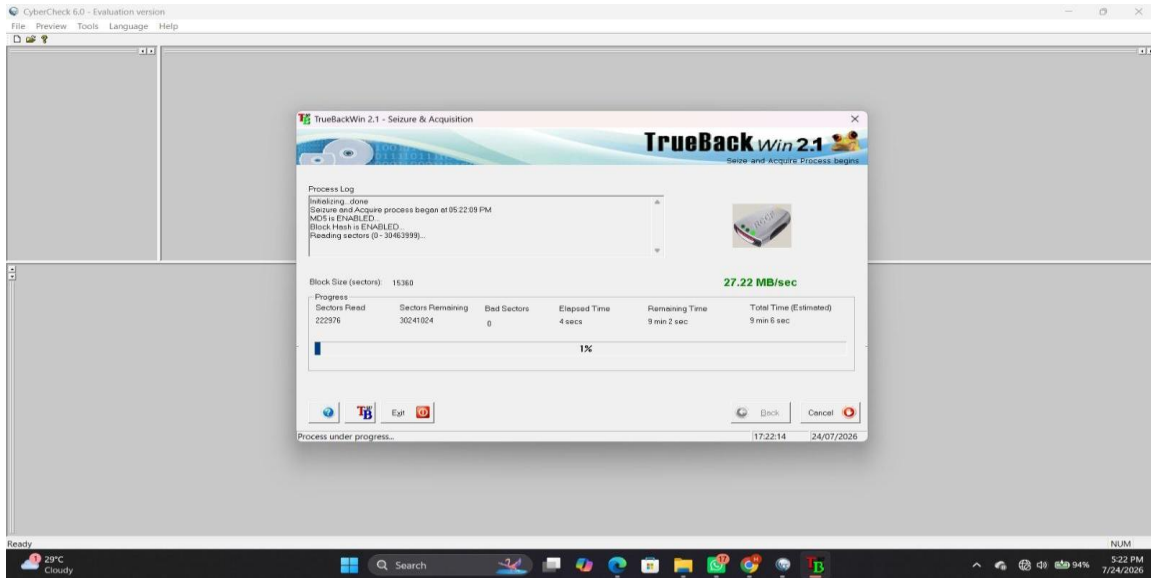


Figure 1.7 — Seizure and Acquisition Execution Window

Step 8: Acquisition Finalization & Evidence Tagging Verification

Upon reaching 100% completion, acknowledge the physical labeling prompt. Safely disconnect the suspect media and affix a physical label containing Exhibit Number ('1'), ID Number ('TBW-020218-005870'), Crime Number ('7'), and Whole-Disk MD5 Checksum ('EFF3-C9E0-72A2-A13E-9582-DA51-3A8B-E653').

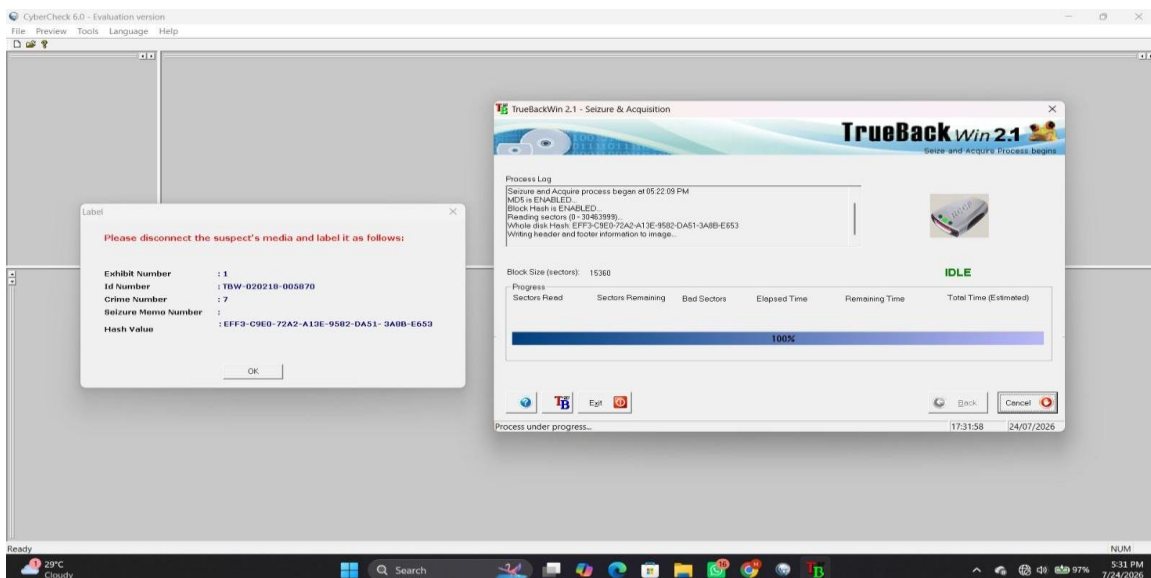


Figure 1.8 — Acquisition Completion and Evidence Labeling Verification Window

Step 9: Seizure & Acquisition Official Audit Report Export

Generate and export the formal 'Seizure & Acquisition Report' via TrueBackWin. Print the document to a virtual PDF driver (Microsoft Print to PDF) to archive a cryptographically signed record of the imaging process for inclusion in court submissions.

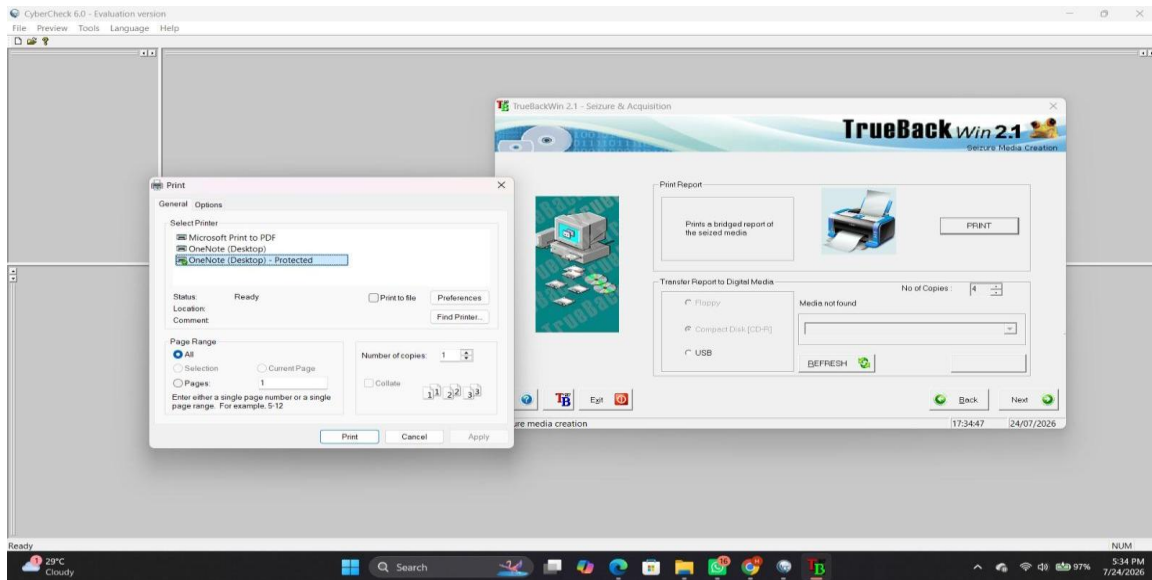


Figure 1.9 — Seizure Report Export and Print Configuration Window

3. Phase II: Deep Analysis & Evidence Carving (CyberCheck Probe)

With the physical bit-stream image created and validated, the investigation proceeds to Phase II: analytical probing, deleted file recovery, file carving, steganography inspection, and signature anomaly detection using CyberCheck Probe.

Step 10: Analyst Authentication & Evidence Mounting

Log into the CyberCheck Probe module. Enter Analyst Credentials (Name: 'Harshal'), input Lab Reference Number ('7'), and select the target forensic bit-stream image path (C:\group 2\7\Pd forensics_01.P01) to mount the evidence.

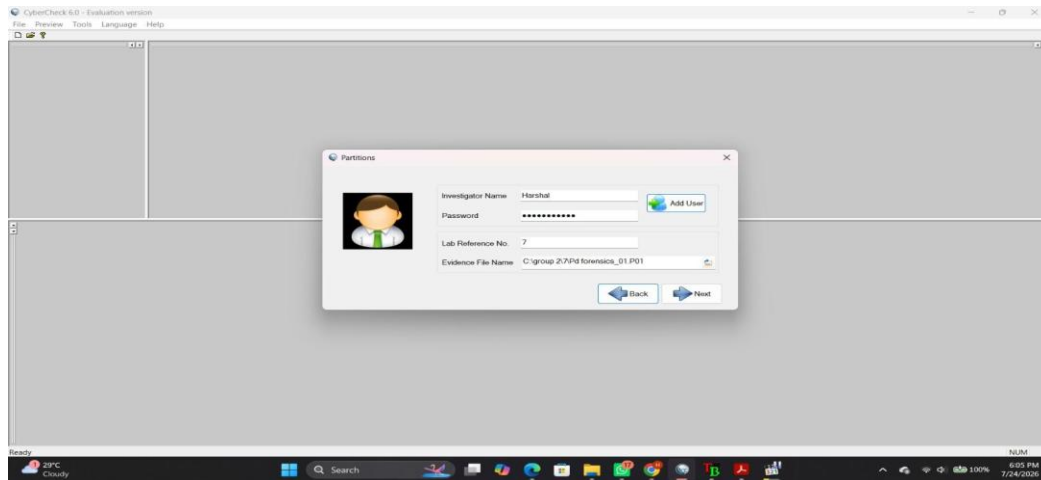


Figure 2.1 — Analyst Authentication and Evidence File Loading Interface

Step 11: Partition Geometry Evaluation & File System Identification

Review the detected volume layout in the Partitions dialog. Select 'Partn_1:' formatted as FAT-32 with a capacity of 14.53 GB and 30,463,968 total sectors. Check the volume selection box to initiate automated file table parsing.

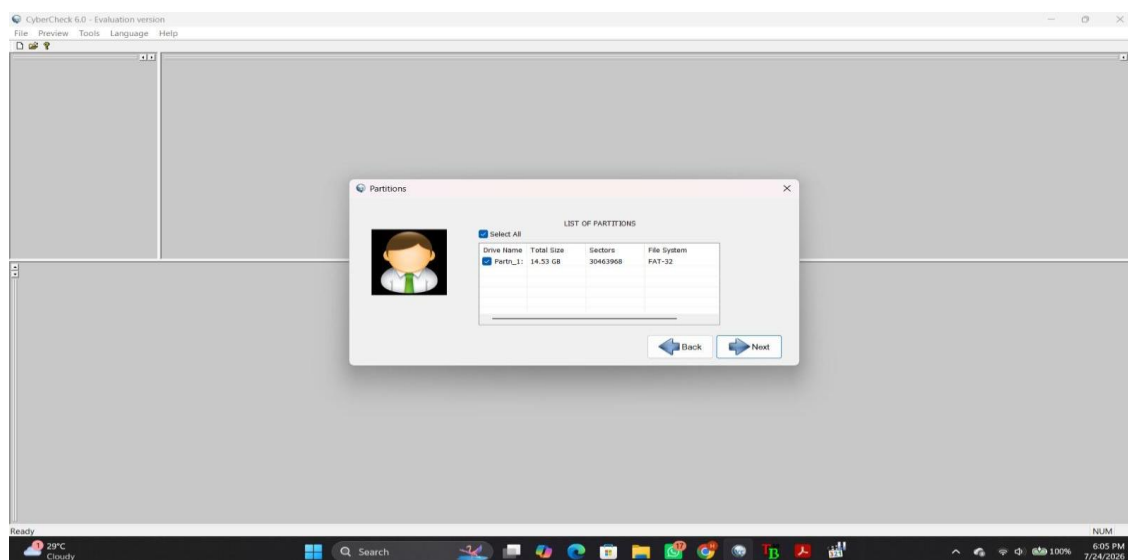


Figure 2.2 — Partition Selection Interface

Step 12: Advanced Probe Analysis & Extraction Settings

Configure analysis parameters: Check 'Enable Hash Verification' to confirm image integrity upon loading, enable 'Extract Used Free Clusters' to carve unallocated drive space, and set the Default Export Folder to E:\ and Temporary Folder to C:\Windows\Temp.

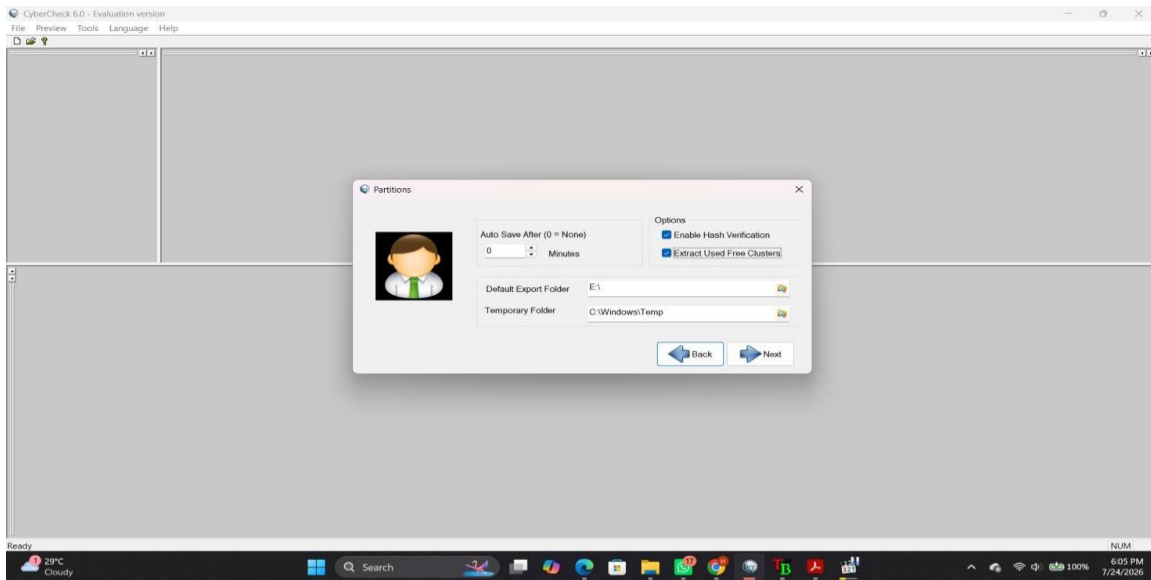


Figure 2.3 — Extraction and Partition Analysis Configuration Options

Step 13: Probe Workspace Initialization & File Table Mapping

Access the primary CyberCheck Probe workspace. The left pane presents the directory tree hierarchy of Partn_1, while the central Table view indexes all existing and deleted directory records. Deleted items are highlighted in red text (e.g., 'IP&MV ALL exp prog...', 'Pushpa_2_The_Rule...', and various Microsoft Word documents).

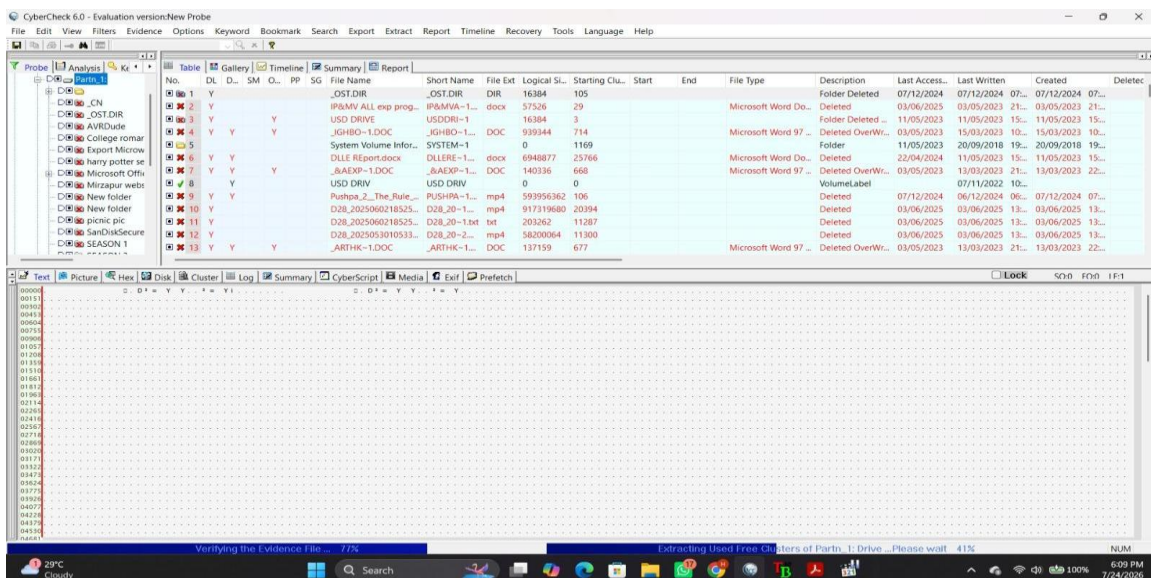


Figure 2.4 — CyberCheck 6.0 Probe Main Workspace and Background Extraction View

Step 14: Background Evidence Verification & Cluster Extraction

Track ongoing background processing threads indicated in the workspace status bar: Evidence File Verification progress (reaching 77% -> 96%) and Used Free Cluster Extraction (reaching 41% -> 49%), ensuring full unallocated space processing for data carving.

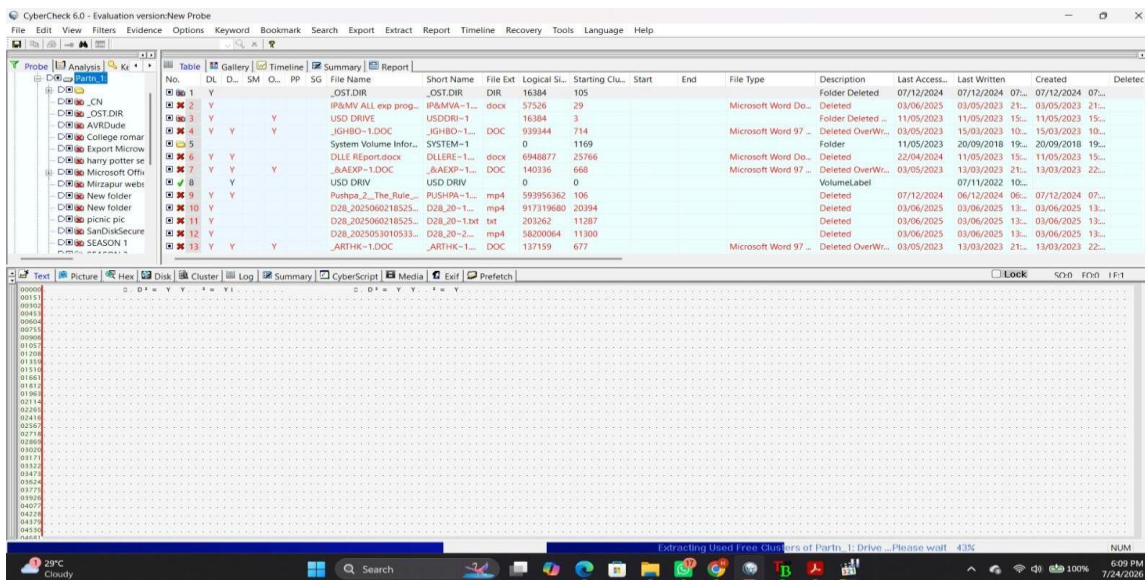


Figure 2.5 — Evidence File Verification and Used Free Cluster Extraction Progress View

Step 15: Options Menu Navigation & Forensic Inspection Tools

Open the 'Options' menu on the top navigation bar. This menu provides access to specialized forensic utility tools: Password Protection Checks, File Signature Analysis, Steganography Detection, Anti-Forensics Detection, and 'Verify Image Hash'.

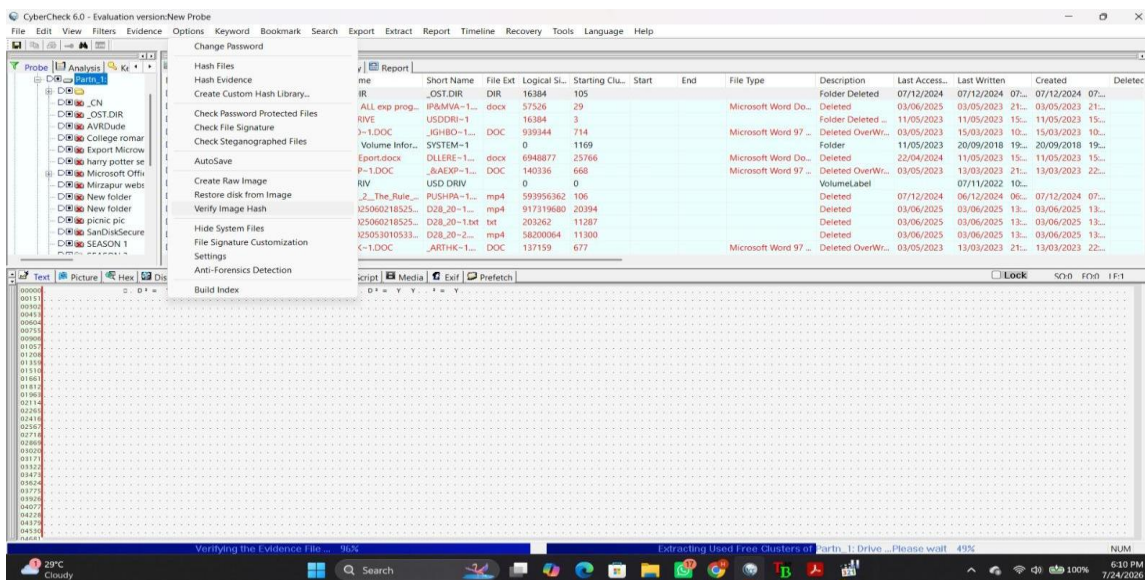


Figure 2.6 — Options Menu and Image Hash Verification Selection Interface

Step 16: Mathematical Image Integrity & Hash Verification

Execute 'Verify Image Hash'. Upon completion, the system displays the 'Hash Success' dialog, confirming that all 1,984 blocks out of 1,984 were verified in 3 minutes 43 seconds, proving zero data alteration occurred since physical acquisition.

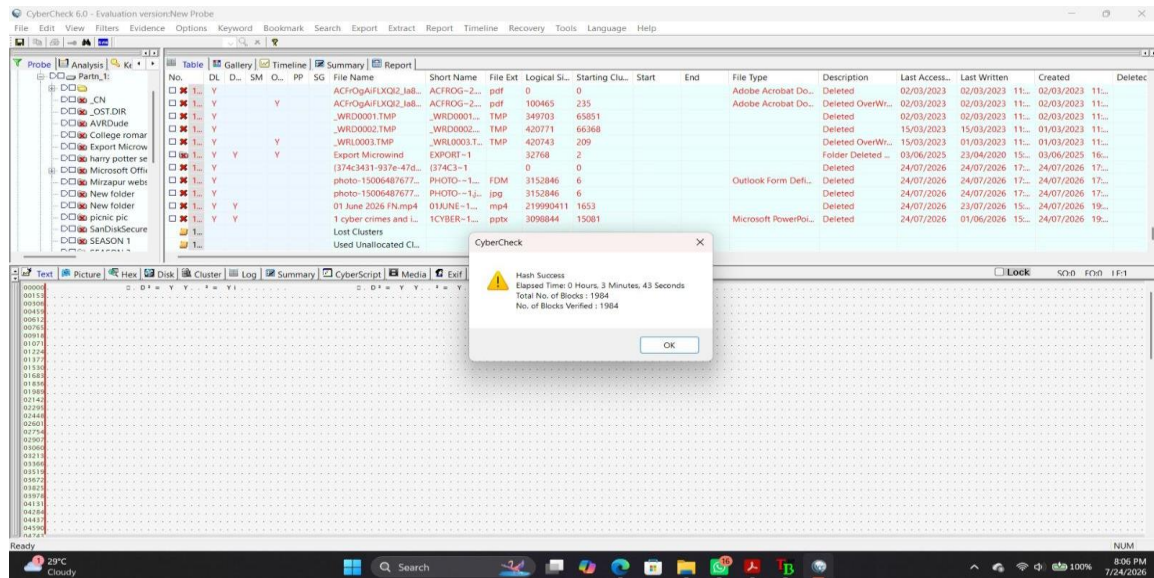


Figure 2.7 — Image Hash Verification Success Notification Window

Step 17: Text & Structural XML Metadata Inspection

Highlight the carved PowerPoint file ('1 cyber crimes and i...', 3,098,844 bytes, starting cluster 15081). Switch to the bottom 'Text' viewer tab to inspect raw ASCII headers and OOXML internal structure, including '[Content_Types.xml' and slide relationship structures ('ppt/slides/ rels/slide1.xml.rels').

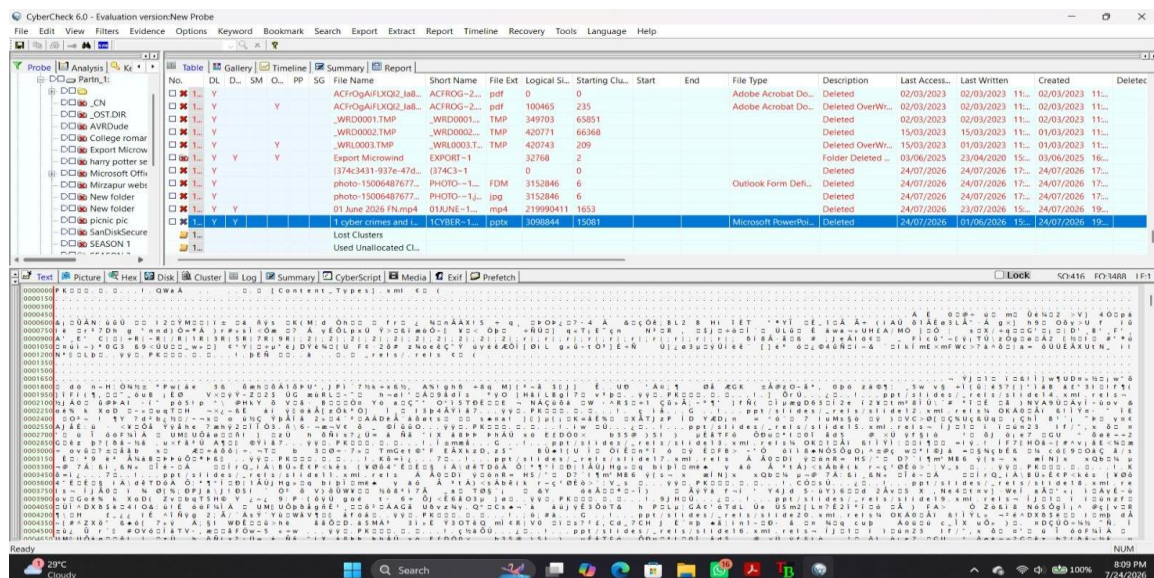


Figure 2.8 — Text Tab Content Inspection for Selected Recovered File

Step 18: Category Filtering & Deleted Partition Reconstruction

Navigate to the 'Deleted Files' category under Partn_1: in the left tree view. Inspect filtered deleted HTML documents (e.g., 'simuOnLayout.htm', 'simuParams.htm', '_TART.HTM'). Simultaneously, monitor background task 'Recovering Deleted Partitions' reaching 31% progress.

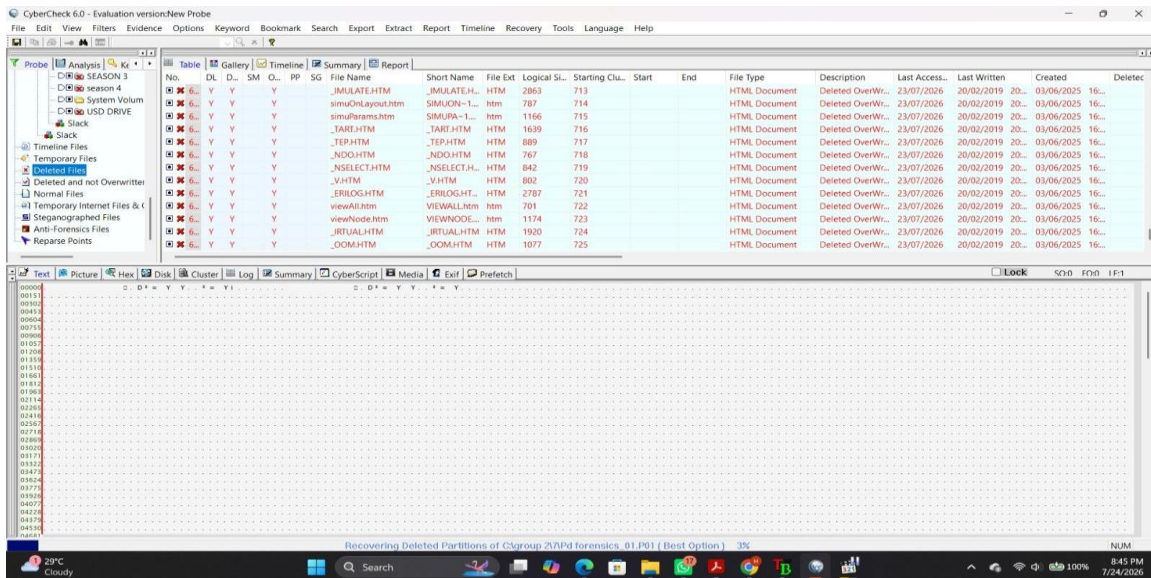


Figure 2.9 — Category Filtering View and Deleted Partition Recovery Progress Window

Step 19: Advanced Keyword & GREP Pattern Search Setup

Open the 'Add Keyword' configuration dialog from the Keyword search tab. Define custom search strings, enable Case Sensitive / Unicode matching, or input regular expressions (GREG) to scan unallocated drive clusters for specific string patterns or file headers.

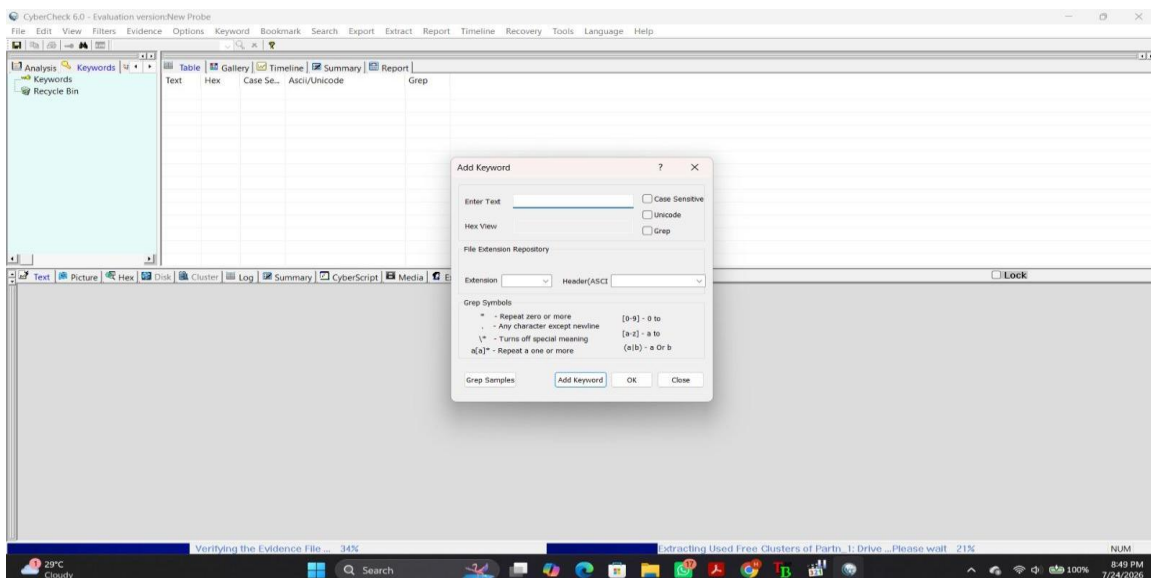


Figure 2.10 — Add Keyword and GREG Search Setup Window

Review search hits for string 'image' (Session1 yielding 8,210 hits). Inspect lower Hex/Text viewer tab to analyze carved system metadata, revealing traces of 'SanDisk SecureAccess V5.0', vault paths ('/app/secureaccess'), and 'EncryptStick' interface strings.



Select 'Check Steganographed Files' from the Options menu. CyberCheck Probe initiates an automated cryptographic scan across all indexed files in Partn 1: to detect embedded hidden data or steganographic cover artifacts.



Step 22: Steganography Analysis Result Evaluation

Evaluate the steganography scan completion prompt: 'Checking for Steganographed Files Completed. Steganographed file(s) exists.' Flagged steganographic files are marked with a 'Y' indicator in the dedicated 'SG' column of the central Table view.

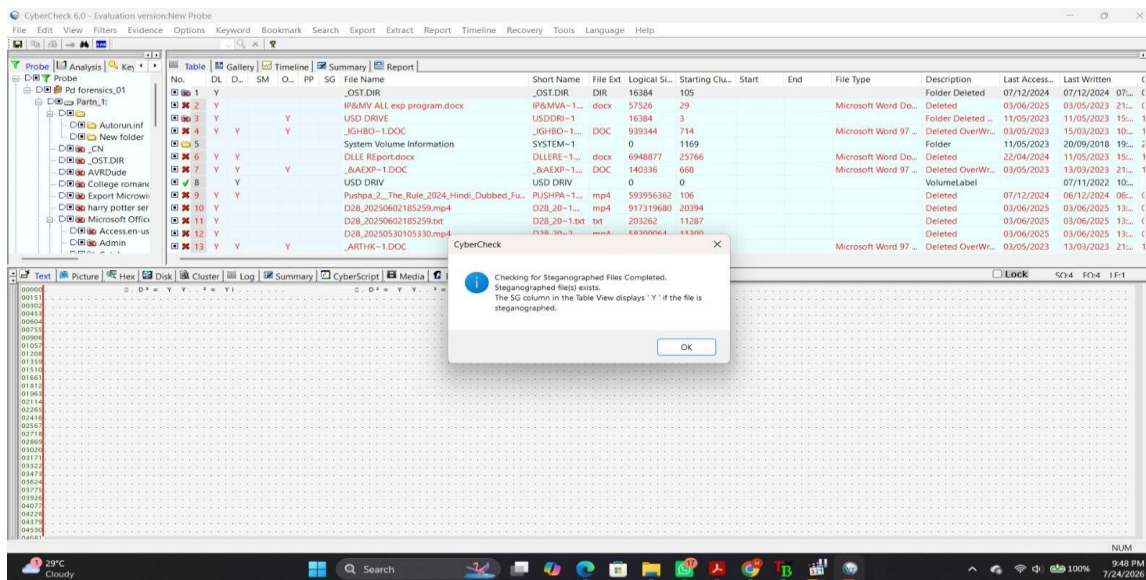


Figure 2.13 — Steganography Check Completion Notification Window

Step 23: File Signature Analysis & Header Verification

Select 'Check File Signature' from the Options menu. CyberCheck performs an automated header-versus-extension comparison across all files to detect extension spoofing or intentional file obfuscation.

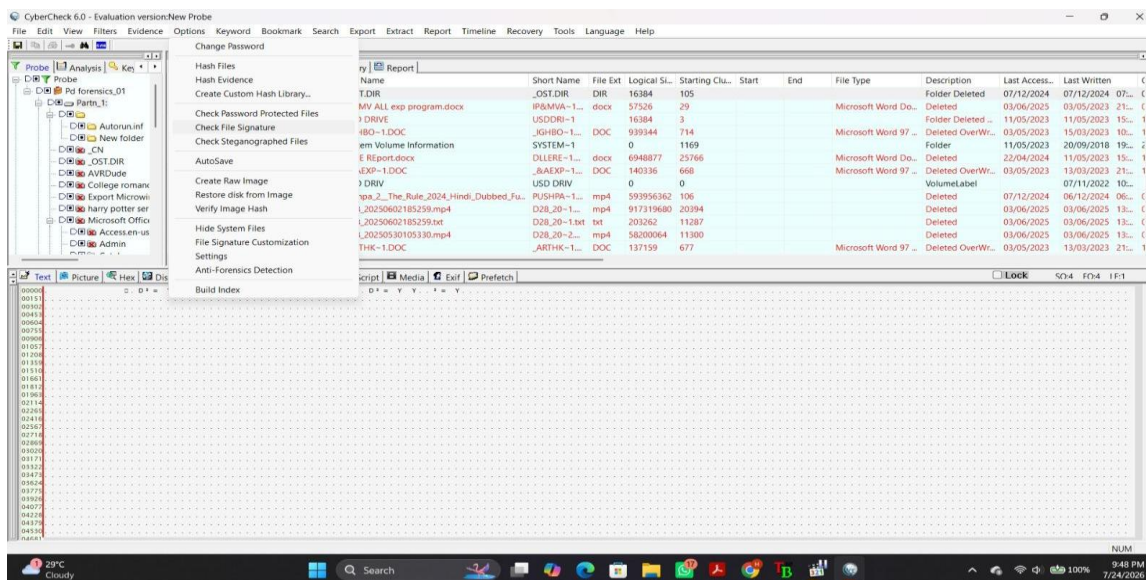


Figure 2.14 — Options Menu and Check File Signature Selection Interface

Step 24: File Mismatch Table Evaluation & Anomaly Identification

Examine the updated Table view columns: 'SM' (Signature Mismatch), 'PP' (Password Protected), and 'SG' (Steganography). Identify files flagged with 'Y' under SM (e.g., '_JGHBO~1.DOC', 'IP&MV ALL exp program.docx'), proving extension tampering.

No.	DL	D.	SM	PP	SG	File Name	Short Name	File Ext	Logical Sl.	Starting Clu.	Start	End	File Type	Description	Last Access.	Last Written
000001	Y					_OST.DIR	_OST.DIR	DIR	16384	105						
000002	Y					IP&MV ALL exp program.docx	IP&MV ALL exp program.docx	docx	57526	29			Microsoft Word Do...	Deleted	03/06/2025	03/05/2023
000003	Y					USD DRIVE	USDRI-1		16384	3				Folder Deleted ...	11/05/2023	11/05/2023
000004	Y					_JGHBO~1.DOC	_JGHBO~1...	DOC	939344	714				Deleted OverWr...	03/05/2023	15/03/2023
000005	Y					System Volume Information	SYSTEM-1		0	1169				Folder	11/05/2023	20/09/2018
000006	Y					DLL Report.docx	DLLRE-1...	docx	6948877	25766				Deleted	22/04/2024	11/05/2023
000007	Y					_AEXP-1.DOC	_AEXP-1...	DOC	140336	668				Deleted OverWr...	03/05/2023	13/03/2023
000008	Y					USD DRIVE	USD DRIV		0	0				VolumeLabel	07/11/2022	10...
000009	Y					Pushpa_2_The_Rule_2024_Hindi_Dubbed_Fu...	PUSHPA-1...	mp4	593956362	106				Deleted	07/12/2024	06/12/2024
000010	Y					D28_20250602185259.mp4	D28_20-1...	mp4	917319680	20394				Deleted	03/06/2025	03/06/2025
000011	Y					D28_20250602185259.mp4	D28_20-1st	txt	202462	11287				Deleted	03/06/2025	03/06/2025
000012	Y					D28_20250530105330.mp4	D28_20-2...	mp4	58200064	11380				Deleted	03/06/2025	03/06/2025
000013	Y					_ARTHC-1.DOC	_ARTHC-1...	DOC	137159	677				Deleted OverWr...	03/05/2023	13/03/2023

Figure 2.15 — File Table View Displaying Mismatched File Signatures and Attributes

Step 25: Visual Evidence Preview in Picture Viewer

Highlight deleted JPEG file ('253abf4f1f4bc16b6d...', size 20,817 bytes, starting cluster 6). Switch to the bottom 'Picture' tab to render and visually verify the carved facial photograph recovered from unallocated space.

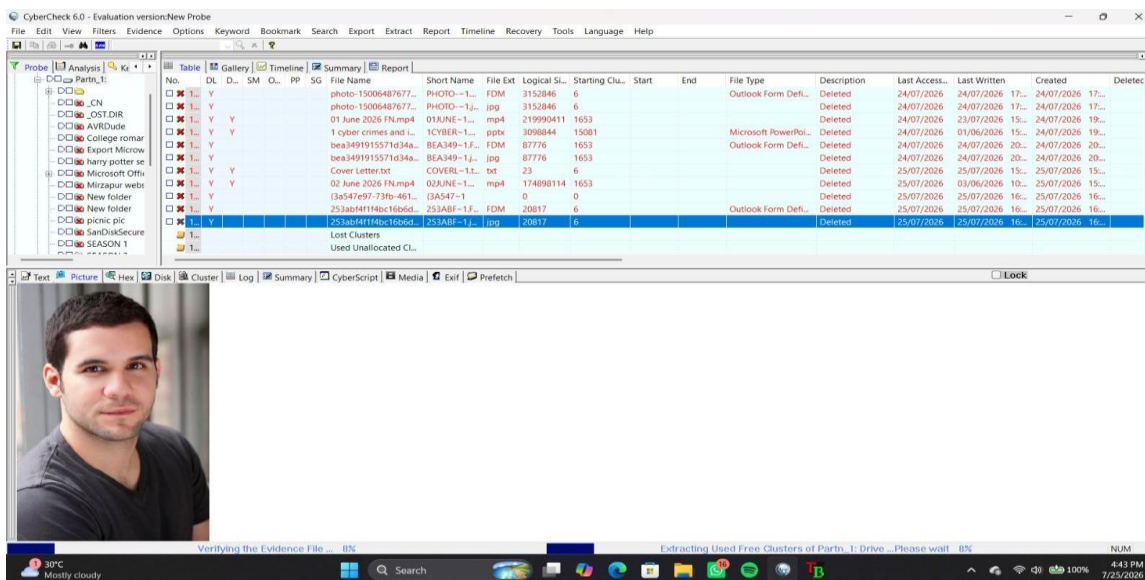


Figure 2.16 — Picture Tab Rendering of Recovered Image File

Step 26: Multimedia Playback & Educational Video Verification

Highlight recovered deleted MP4 video file ('02 June 2026 FN.mp4', size 174,898,114 bytes, starting cluster 1653). Switch to the bottom 'Media' tab to execute real-time playback, rendering lecture slides on 'Digital Evidence Seizure' alongside embedded video.

The screenshot displays the CyberCheck 6.0 interface. The top menu bar includes File, Edit, View, Filters, Evidence, Options, Keyword, Bookmark, Search, Export, Extract, Report, Timeline, Recovery, Tools, Language, and Help. The main window is divided into several panes. On the left, a 'Probe' pane shows a tree view of the file system. The central pane displays a table of recovered files. The bottom pane is the 'Media' tab, which contains a video player and a slide titled 'Seizure'.

No.	DL	D.	SM	CL	PP	SG	File Name	Short Name	File Ext	Logical SL	Starting Clu.	Start	End	File Type	Description	Last Access...	Last Written	Created	Deletec
1	Y						photo-15006487677...	PHOTO-1...	FDM	3152846	6				Outlook Form Defi...	Deleted	24/07/2026	24/07/2026	17...
2	Y						photo-15006487677...	PHOTO-1...	jpg	3152846	6				Deleted	24/07/2026	24/07/2026	17...	
3	Y	Y					01 June 2026 FN.mp4	01JUNE-1...	mp4	219990411	1653				Deleted	24/07/2026	23/07/2026	15...	
4	Y						1 cyber crimes and L...	1CYBER-1...	pptx	3098844	15081				Microsoft PowerPol...	Deleted	24/07/2026	01/06/2026	15...
5	Y						bea3481915571d34a...	BEA348-1...	FDM	87776	1653				Outlook Form Defi...	Deleted	24/07/2026	24/07/2026	20...
6	Y						bea3481915571d34a...	BEA348-1...	jpg	87776	1653				Deleted	24/07/2026	24/07/2026	20...	
7	Y	Y					Cover Letter.txt	COVERL-1...	txt	23	6				Deleted	25/07/2026	25/07/2026	15...	
8	Y	Y					02 June 2026 FN.mp4	02JUNE-1...	mp4	174898114	1653				Deleted	25/07/2026	03/06/2026	10...	
9	Y						(3a547e97-73fb-461...	(3A547-1		0	0				Deleted	25/07/2026	25/07/2026	16...	
10	Y						253ab04f114bc16b6d...	253ABF-1...	FDM	20817	6				Outlook Form Defi...	Deleted	25/07/2026	25/07/2026	16...
11	Y						253ab04f114bc16b6d...	253ABF-1...	jpg	20817	6				Deleted	25/07/2026	25/07/2026	16...	
12	Y						Lost Clusters								Deleted	25/07/2026	25/07/2026	16...	
13	Y						Used Unallocated CL...								Deleted	25/07/2026	25/07/2026	16...	

The 'Media' tab shows a video player with a 'Seizure' slide. The slide text reads: 'Seizure is the process of capturing suspect computer or storage media for evidence collection and generating a hash value corresponding to the evidence Seized.' Below the text are two images: a yellow evidence bag and a computer monitor displaying a video. The video player has a 'Copilot' button and a 'NUM' indicator.

Figure 2.17 — Media Tab Video Playback and Evidence Verification Interface

4. Forensic Analytical Comparison & Evaluation

To evaluate the technical capabilities of CyberCheck 6.0 relative to other digital forensic suites (such as Foremost, Scalpel, Bulk_Extractor, X-Ways Forensics, and Magnet AXIOM), a comparative matrix was synthesized based on experimental performance during evidence recovery:

Forensic Feature / Metric	CyberCheck 6.0	Foremost / Scalpel	X-Ways Forensics	Magnet AXIOM
Seizure & Chain of Custody	Integrated CoC Form & Native Printing	None (CLI Tool)	Case Metadata Logging	Full Case Details & Exporter
Physical Bit-Stream Imaging	TrueBackWin Split Raw (.P01)	Requires dd / FTK Imager	Raw .dd / .E01 Format	Acquire E01 / Raw Disk
Cryptographic Verification	MD5 & Block Hash (15360 Sectors)	External Hash Utilities	MD5 / SHA-1 / SHA-256	Automated Hash Verification
Steganography Detection	Automated Native Scanning (SG Column)	None	Manual Third-party Plugin	Artifact-based Detection
Signature Mismatch Analysis	Automated Header/Extension Check (SM)	Basic Carving Header/Footer	Comprehensive Type Audit	Automated Category Classification

5. Conclusion & Recommendations

This minor project successfully demonstrated a complete forensic investigation workflow using the CyberCheck 6.0 suite. TrueBackWin 2.1 proved highly efficient in performing bit-stream physical imaging with dual MD5 and block-level hashing, ensuring strict write-blocking compliance and legal chain of custody documentation. Subsequent probing within CyberCheck Probe validated complete mathematical image integrity (1,984 blocks verified), successfully carved deleted presentation files, recovered unallocated multimedia artifacts (JPEG photo and MP4 video), identified steganographic content, and flagged extension spoofing via signature mismatch analysis.

Key Recommendations for Future Investigations:

1. Always maintain physical and software write-blockers during initial media connection.
2. Utilize dual hash algorithms (MD5 and SHA-256) to eliminate theoretical collision risks.
3. Cross-validate carved artifacts using multi-tool comparative approaches (e.g., combining CyberCheck with X-Ways or Magnet AXIOM).

6. References & Standards

1. ISO/IEC 27037:2012 — Guidelines for identification, collection, acquisition, and preservation of digital evidence.
2. C-DAC Thiruvananthapuram — CyberCheck 6.0 & TrueBackWin 2.1 Technical Documentation and Operational Manuals.
3. Digital Forensic Tool Testing (DFTT) — Standard Evaluation Datasets and Test Images.
4. Carrier, B. (2005). File System Forensic Analysis. Addison-Wesley Professional.
5. Casey, E. (2011). Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet. Academic Press.

Appendix A: Seizure & Acquisition Technical Audit Report

Page 1 of 1

Seizure & Acquisition Report

Report generated by TrueBackWin version 2.1 developed by CDAC, Thiruvananthapuram

Investigator's Name : group 2
Investigator's Rank : 1st
Police Station : Mumbai police Station
Crime Number : 7
Seizure Memo Number :
Lab Ref Number : 123
Place of Seizure : Mumbai
Date of Hashing : 24/07/2026
Time of Hashing : 05:20:37 PM
Suspect's Name : Rahul
Suspect's Address1 : Juhu mumbai
Suspect's Address2 : Nashik
Name of Witness1 : Harshal
Address1 of Witness1 : mumbai
Address2 of Witness1 : mumbai
Name of Witness2 : Akshay
Address1 of Witness2 : mumbai
Address2 of Witness2 : mumbai
Evidence File Name : Pd forensics
Start Time : 05:22:09 PM
End Time : 05:32:38 PM

Processor Details

Processor Name : Intel(R) Core(TM) i7-10870H
Speed : 2.211Ghz

Exhibit Details

Exhibit Number : 1
Disk Type : USB STORAGE
Model Number : SanDisk
Serial Number : 4C530001170403112591
Image Format : TrueBackWin Image
Hash Type : MD5
Total Sectors : 30464000
Bytes per Sector : 512
Block Size : 15360 sectors
Number of Blocks : 1984

Hash Values

Hash Value of Whole Media : EFF3-C9E0-72A2-A13E-9582-DA51-3A8B-E653
Hash Value of Block Hashes : 4D49-D69E-E810-4DB4-C605-E711-A4F4-36B1
Hash Value of Report Data : E88B-3CAC-84A8-360F-6B8E-2D34-D7DD-D2FA

Appendix B: CyberCheck Probe Investigation Case Report

Page 1 of 6

CyberCheck Report

Probe : Untitled

Evidence Name : Pd forensics_01.P01

File C:\group 2\7\Pd forensics_01.P01 was acquired by group 2,1st at 25/7/2026 16:33:5

Evidence acquired using TrueBackWin version 2.0

During acquisition the suspect's media was Write Protected

Crime Number : 7
Lab Reference Number : 123
Police Station : Mumbai police station
Acquisition Notes : Forensics
Media Type : USB Storage Device

Drive Geometry :

Total Capacity	14.53 GB
Total Sectors	30464000
Bytes per Sector	512
Cylinders	56065
Heads	1
Sectors per Track	63

File Integrity :

Acquisition Hash (MD5) : 127A-0965-DA9B-EE93-AEFD-31DF-EBBC-E314
Hash Verification Status : Verified and successful.

Partition Details :

Volume Partn_1: Parameters			
Volume Label	NO NAME	Volume Serial	14C8-03BA
File System	FAT-32	Partition Type	0C
Total Capacity	14.53 GB	Total Sectors	30463968
Number of Folders	41	Number of Files	205
Sectors per Track	63	Cluster Size	16384
Sectors per Cluster	32	Free Clusters	914939
Number of FATs	2	Root Sector	14913
Sectors per FAT	7434	Heads	255

Logged in by Harshal with Lab Reference Number 7 on Saturday, July 25, 2026 at 04:43:12 PM

Report Data 1**File Details**

File Name : 1 cyber crimes and info security.pptx
Short Name : 1CYBER~1.pptx
File Extension : pptx
Logical Size : 3098844
Starting Cluster : 15081
Signature : Microsoft PowerPoint Presentation
Description : Deleted
Last Accessed Date : 24/07/2026
Last Written : 01/06/2026
Created : 24/07/2026
File Deleted : Yes
Full Path : Pd forensics_01\Partn_1:\1 cyber crimes and info security.pptx
Slack Size : 14116
Cluster List : 15081 - 15270

Content :-

[Click to view file content](#)

Comment :- Deleted ppt Found

Appended on Saturday, July 25, 2026 04:58:18 PM

Report Data 2**File Details**

File Name : Cover Letter.txt
Short Name : COVERL~1.txt
File Extension : txt
Logical Size : 23
Starting Cluster : 6
Signature :
Description : Deleted
Last Accessed Date : 25/07/2026
Last Written : 25/07/2026
Created : 25/07/2026
File Deleted : Yes
Full Path : Pd forensics_01\Partn_1:\Cover Letter.txt
Slack Size : 16361
Cluster List : 6

Content :-

[Click to view file content](#)

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Report Data 3

File Details

File Name	: 02 June 2026 FN.mp4
Short Name	: 02JUNE~1.mp4
File Extension	: mp4
Logical Size	: 174898114
Starting Cluster	: 1653
Signature	:
Description	: Deleted
Last Accessed Date	: 25/07/2026
Last Written	: 03/06/2026
Created	: 25/07/2026
File Deleted	: Yes
Full Path	: Pd forensics_01\Partn_1\02 June 2026 FN.mp4
Slack Size	: 1086
Cluster List	: 1653 - 12327

Content :-

[Click to view file content](#)

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Report Data 4

File Details

File Name	: 253abf4f1f4bc16b6dc04571f8d21624.jpg
Short Name	: 253ABF~1.jpg
File Extension	: jpg
Logical Size	: 20817
Starting Cluster	: 6
Signature	:
Description	: Deleted
Last Accessed Date	: 25/07/2026
Last Written	: 25/07/2026
Created	: 25/07/2026
File Deleted	: Yes
Full Path	: Pd forensics_01\Partn_1\253abf4f1f4bc16b6dc04571f8d21624.jpg
Slack Size	: 11951

Cluster List : 6 - 7

Picture :-



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Report Data 5

File Details

File Name	: Files.exe
Short Name	: FILES.exe
File Extension	: exe
Logical Size	: 96256
Starting Cluster	: 120
Signature	: Application
Description	: Deleted
Last Accessed Date	: 11/04/2019
Last Written	: 02/02/2014

Created : 08/04/2019
File Deleted : Yes
Full Path : Pd forensics_01\Partn_1:\Files.exe
Slack Size : 2048
Cluster List : 120 - 125

Content :-

[Click to view file content](#)

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Report Data 6

File Details

File Name : photo-1500648767791-00dcc994a43e.jpg.fdmdownload
Short Name : PHOTO~1.FDM
File Extension : FDM
Logical Size : 3152846
Starting Cluster : 6
Signature : Outlook Form Definition
Description : Deleted
Last Accessed Date : 24/07/2026
Last Written : 24/07/2026
Created : 24/07/2026
File Deleted : Yes
Full Path : Pd forensics_01\Partn_1:\photo-1500648767791-00dcc994a43e.jpg.fdmdownload
Slack Size : 9266
Cluster List : 6 - 198

Content :-

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Report Data 7

File Details

ACFrOgAiFLXQI2_la8HaLGwL62nF-
File Name : TJ34DHNZ2MPDO3CRHI6v5SDZeTtMtnpKZVurf1cpBimFslrFbdAF_hfMM-
XQnpS5KjS9nLftKuuLEQTUFuXXSYCZjnuujZi7VMVMkXRCafgBSgEjRv_nJF.pdf
Short Name : ACFROG~2.pdf
File Extension : pdf
Logical Size : 100465
Starting

Cluster : 235
Signature : Adobe Acrobat Document
Description : Deleted OverWritten
Last
Accessed : 02/03/2023
Date
Last Written : 02/03/2023
Created : 02/03/2023
File Deleted : Yes
Pd forensics_01\Partn_1:\ACFrOgAiFLXQl2_la8HaLGwL62nF-
Full Path : TJ34DHNZ2MPDO3CRHi6v5SDZeTtMtnpKZVurf1cpBimFslrFbdAF_hfMM-
XQnpS5KjS9nLftKuuLEQTUFuXXSYCZjnuujZi7VMVMkXRCafgBSgEjRv_nJF.pdf
Slack Size : 14223
Cluster List : 235 - 241

Content :-

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